D06 · Cold Chain & Specialized Logistics

01 · Cold chain design, temperature monitoring & compliance

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Cold chain is the controlled-temperature system — packaging, equipment, monitoring, procedures and qualified partners — that keeps pharmaceuticals, biologics and perishable foods within their specified range from origin to delivery, with documented evidence that it did.

2 WHY IT MATTERS

A temperature excursion can destroy product value or create a patient-safety risk, and the damage is often invisible to the eye. **For regulated products, the record is part of the product**: in Mexico, good-practice rules for medicines (NOM-059-SSA1) and sanitary oversight by COFEPRIS require controlled storage and documented handling of deviations. A cold chain without data is a cold chain you cannot defend in an audit.

3 HOW IT IS DONE

1. **Define the range and the risk per product.** Storage condition, stability data and allowable excursions determine packaging and mode.
2. **Qualify the whole chain.** Warehouses, docks, vehicles, packaging and partners — not only the carrier.
3. **Monitor continuously, alert before the limit.** Data loggers with pre-limit thresholds and, for high-value flows, real-time transmission.
4. **Investigate every excursion formally.** Impact assessment, disposition (release, quarantine, reject) and CAPA, documented.

IN PRACTICE

Move from passive to active monitoring for high-value or high-risk products

Passive loggers only tell you that something went wrong after the damage is done.

Treat every excursion as a formal quality event

Documented investigation, impact assessment and CAPA are required both for compliance and for learning.

Qualify the entire chain, not just the carrier

Warehouse docks, staging areas, and receiving processes are frequent sources of excursions that never appear in transit data.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Pharmaceutical distributor, Mexico — Previously used passive loggers read only at destination. Several excursions were discovered after product had already been received into inventory.

Element	Before	After
Monitoring	Passive loggers	IoT loggers with pre-limit alerts
Time to detect excursion	At destination (often 24–72 h later)	Average 47 minutes from breach
Product loss from excursions	Baseline	-61% in 12 months
Audit findings related to temperature control	Recurring	Zero major findings in subsequent COFEPRIS inspection

Result: Real-time visibility allowed intervention (re-icing, re-routing, or accelerated delivery) before product was compromised. The compliance posture improved materially.

HOW IT IS MEASURED

Temperature Excursion Rate %

(Shipments with confirmed excursion / Total cold-chain shipments) × 100

Benchmark: <0.5% for mature pharma cold chains · higher rates require root-cause program

Mean Time to Detect Excursion

Average time from temperature limit breach to system alert / human awareness

Benchmark: <1 hour for real-time systems · days for passive loggers

WHAT YOU WOULD USE

Cold Chain Monitoring

Sensitech / Controlant / Berlinger / Roambee — Leading IoT temperature monitoring platforms with real-time alerts and compliance reporting.

Integration with QMS / Track & Trace systems — Excursion data should flow into the quality management system for formal investigation and disposition.

4 SO WHAT

Design for evidence, not just for cold. Qualify every link, alert before the limit, and treat every excursion as a quality event with a documented decision.

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